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Problem 2: AI Lawyer for India | Consumer grievance companion
Scoped, auditable and demonstrable. This is an information and drafting tool, not legal representation.

01 Problem and why this scope

For a small online-purchase dispute, a buyer often struggles to organise dates, proof and a written grievance. I deliberately focus on **refunds, defective purchases and failed deliveries** rather than pretending to know all Indian law. The user leaves with a reviewed case summary, a seller complaint draft and official escalation links.

02 What actually works

• **Chat-first intake:** English/Hindi text conversation with sequential follow-up questions; extracts only clearly stated amounts, dates and issue cues.

• **Case workspace:** eight editable fields (seller, item, amount, date, issue, prior contact, evidence, requested remedy); users confirm before drafting.

• **Useful output:** downloadable grievance letter and JSON case record; evidence checklist, verified links to the National Consumer Helpline and e-Jagriti.

• **Safety:** unfamiliar problems are not silently classified as consumer cases; urgent court/criminal/safety terms and known non-consumer disputes trigger refusal/escalation.

• **Optional AI and voice:** a swappable OpenAI-compatible adapter suggests JSON facts only (explicit user acceptance needed). Voice can be locally transcribed with faster-whisper. Neither feature is required to run the free default app.

03 Architecture and model boundaries

INPUT	INTAKE + TRIAGE	CONFIRM	ACTION
Chat / optional local voice	Rule-based extraction; optional LLM suggests facts	User edits eight fields	Grievance TXT, case JSON, NCH / e-Jagriti

04 Source grounding and assumptions

No LLM generates legal cases or references. FAQ answers are fixed and attached to official sources: India Code (Consumer Protection Act, 2019), Department of Consumer Affairs (NCH, 1915), and e-Jagriti. The default app requires no API or account. A model can only suggest factual fields; all factual claims need user confirmation. No compensation, legal outcome, forum, filing fee or case-specific deadline is promised. No real complaint is automatically submitted.

05 How to run and reproduce

Windows: `python -m pip install -r requirements.txt`; then `python -m streamlit run app.py`. **Tests:** `python -m unittest discover -s tests -v`. Synthetic evidence: `test_cases.json` and `raw_outputs/T01-T12.json`. Deployment steps in `DEPLOYMENT.md`.

VALIDATION, SAFETY & SCALE

06 Quality checks and raw runs

15 deterministic engine checks pass (scope, Hindi, missing fields, guarded drafts, source mapping and redaction). **12 synthetic raw outputs** were generated without hand-editing JSON. This proves code-path behaviour, not real legal correctness or a live browser test.

• **T01 standard refund:** completed fields yield grievance; **T03 Hindi:** Hindi factual complaint.

• **T04 missing proof/amount:** no invented fields or draft; **T05 tenancy:** referral outside scope.

• **T06 court summons:** blocks automated drafting; **T07 fabricated precedent request:** does not invent authority or guarantee a remedy.

• **T09 ambiguous issue:** remains uncertain; **T10 Hindi court matter:** escalates; **T12 defective jacket:** replacement request is rendered.

07 Honest limitations and failures

Rules are language-limited and can miss slang or veiled urgency; conversational replies can be misclassified as fields, dates and sellers are not verified, and user-entered false facts can appear in a draft. Hindi free-text comprehension remains partial. Voice requires a local download and may mis-transcribe. The optional external LLM can still hallucinate extracted facts, so suggestions require acceptance and cannot author legal positions. **Known design failure:** a broad “AI lawyer” concept would overpromise law coverage; I narrowed scope and locked legal assertions to checked sources.

08 AI use, safety and human-in-the-loop

ChatGPT was used to help implement the code, test plan, interface and documentation. The primary free workflow is deterministic rather than pretending to be a general legal LLM. An optional external model is plug-and-play, but not required to reproduce the results. A user checks the facts and draft before using it; an actual lawyer must advise on court matters, criminal allegations, filing deadlines, jurisdiction, contested entitlement or urgent issues. Sensitive identifiers should not be entered; any external model receives partially redacted text only when the user opts in.

09 Scale and success metrics

The input and output contract is a stable Python `Case` record. For scale: stateless hosting, source-update review, regression tests across languages, versioned prompts, optional API-routing and lawyer escalation. Evaluate on real consenting users: completion rate, corrections per case, time to usable draft, safe-refusal recall, unsupported legal claim rate (target zero), and successful navigation to the official grievance channel. These product outcomes are **proposed metrics, not measured results**.

10 Sources & submission

Official sources checked 08 Nov 2026: indiacode.nic.in (Consumer Protection Act 2019); consumeraffairs.gov.in/pages/nch (1915); consumerhelpline.gov.in; e-jagriti.gov.in; nalsa.gov.in. Submit this **two-page PDF**, a working app or GitHub link, and the raw-outputs folder. Optional walkthrough is up to three minutes. A hosted URL must be created using the owner’s hosting account.